

JavaScript mid-term exam (2021-22-1)

Due No Due Date Points None

Important information

- You have **120 minutes** to complete the exam. We provide you **additional 30 minutes** for filling in the `README.md` (see template below), reading the tasks, downloading the materials, compressing and uploading your solution.
- Solutions must be uploaded to ELTE Canvas. **The system closes at 18:30 (Hungarian time) and there will be no opportunity to hand in your solution after the aforementioned time.**
- You may use **any kind of material** (including documentation, lectures, class exercises, cheat sheets, etc) during the exam, **except human help** (synchronous, asynchronous, chat, forums, etc)!
- The tasks are independent and can be solved in **any order**.
- To begin your work you should first [download the starting package we have prepared](#). Each task has its own directory in the package. All tasks are somewhat prepared. Generally, you only have to edit the provided `.js` files, but feel free to divide up your solution into multiple `.js` files, or even edit the HTML if needed.
- The starting package contains a template called `README.md` where you should fill in your name and Neptun code (the `<>` characters must be removed). **Submissions without a properly filled `README.md` file will not be graded!**
- Each directory contains a file named `TASKS.md`. Replace the space in `[ ]` with an `x` where you have (even partially) succeeded to complete the task. This will make us easier to find your solution when we grade your exam.

1. task: Weather JSON (8 points)

A weather API JSON response is given. Solve the following tasks with this JSON. *Technical help: From subtask b. it is recommended to solve the problems with array methods.*

- a. (1 point) The JSON is specified as a text in a multiline text input field (textarea). Click the "Calculate statistics" button to read the contents of the textarea, and convert the JSON string to a JavaScript object, and write it to the console! *Note: Additional tasks will work with the JavaScript object that was read this way. If you cannot convert it from the JSON string, simply copy it from the textarea field and assign to a variable. In this case, there is no point for this subtask.*
- b. (2 point) Write the smallest value of daily maximums in the item with the identifier `task2`. You have to work with the `daily` array, the daily maximum is the `max` property of `temp`.
- c. (2 point) Write in the item with the id `task3` whether the air pressure was above 1010 every day! The `daily` array must be used, the air pressure is included in the `pressure` property.
- d. (2 points) Write the average daily temperature in the item with the identifier `task4`! You have to work with the `daily` array, the daily felt temperature is the `day` property of the `feels_like` record.
- e. (1 point) Write the number of windy hours in the item with the identifier `task5`! You need to work with the `hourly` array, and a windy hour is when the `wind_speed` is greater than 3.

WEATHER JSON

```
{
  "lat": 47.45,
  "lon": 19.03,
  "timezone": "Europe/Budapest",
  "timezone_offset": 7200,
  "hourly": [
    {
      "dt": 1603094400,
      "temp": 9.28,
      "feels_like": 7.23,
      "pressure": 1026,
      "humidity": 87,
      "dew_point": 7.23,
      "clouds": 0,
      "visibility": 10000,
      "wind_speed": 2.01,
      "wind_deg": 277,
      "weather": [
        {
          "id": 800,
          "main": "Clear",
          "description": "clear sky",
          "icon": "01d"
        }
      ],
      "pop": 0
    },
    {
      "dt": 1603098000,
      "temp": 10.39,
```

Calculate statistics

The smallest maximum temperature value of the days

12.3

The air pressure is always above 1010

true

The average "feels like" daily temperature

10.49625

Number of windy hours

12

2. task: Image filters (8 points)

In CSS it is possible to apply different filters to an image: e.g. `filter: contrast(200%) hue-rotate(90deg);`. If there are multiple filters, they must be separated by a space. In this task you are given an image and a couple of filters. We want to see what effect the filter has on the image: if you check the checkbox of a filter, apply it on the image and use the slider next to it to adjust the value of the filter. Read the technical help at the end of the task!

- a. (2 points) If the checkbox of a filter is active, read the filter text specified as the `value` of the checkbox, and replace the `#` character with the value of the slider after the checkbox. Apply the resulting filter text to the style attribute of the image!
- b. (2 points) If more than one check box is checked, apply each one to the image!
- c. (1 point) If you uncheck a checkbox, that filter will not affect the image!
- d. (2 points) If you change the value of a slider, and the checkbox next to it is clicked, apply the new slider value to the image!
- e. (1 point) A `filters` array is given in `index.js`. When loading the page, use it to generate each filter (`<label>` and its contents), replacing the static ones (checkbox `value`, label text, slider `min`, `max`, `value`)!

Technical help: both turning the checkbox on and off and moving the sliders generate an `input` event. Thus, the task can be solved with only one event handler listening on the `div` containing the checkboxes. In this event handler select the checked checkboxes, read the value of those checkboxes and the values of the corresponding sliders, create the filter text, and apply the resulting image filter text to the style of the image.

IMAGE FILTERS



FILTERS

☐ Blur <input type="range"/>	☐ Brightness <input type="range"/>	☐ Contrast <input type="range"/>
☐ Grayscale <input type="range"/>	☒ Hue rotate <input type="range"/>	☐ Invert <input type="range"/>
☐ Opacity <input type="range"/>	☐ Saturate <input type="range"/>	☐ Sepia <input type="range"/>

3. task: Timeline (8 points)

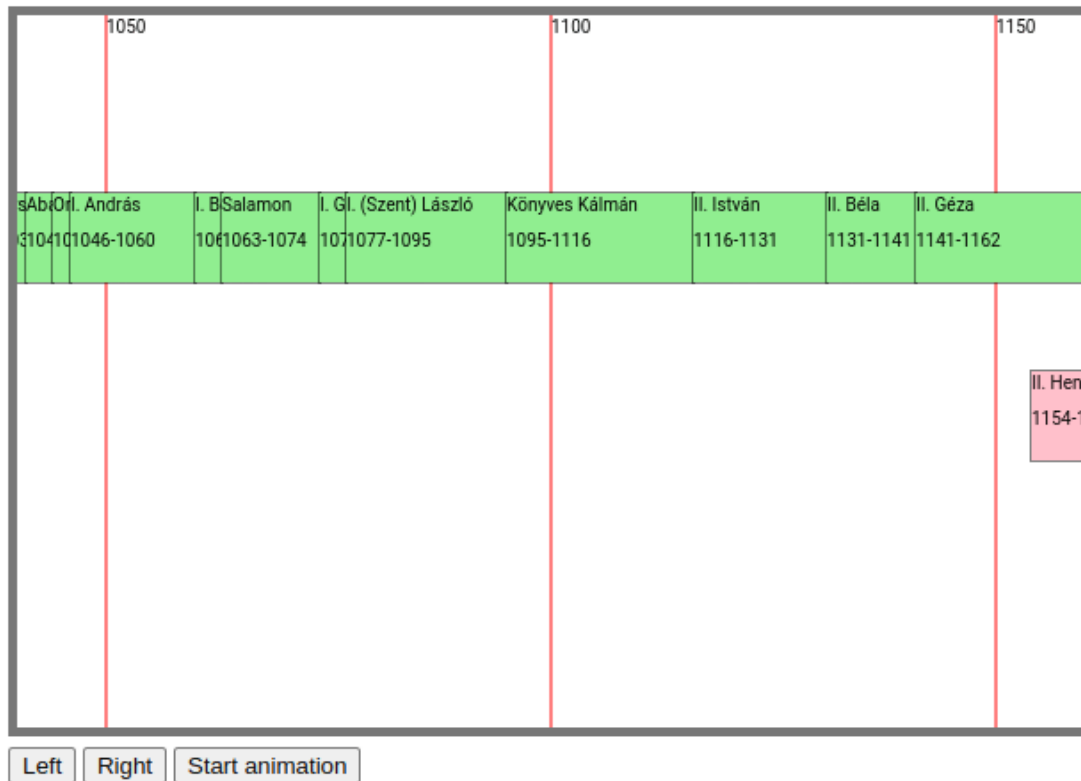
We would like to display the intervals of the reign of Hungarian and English kings in Canvas. For this, `kings.js` provides the appropriate information for the kings of the House of Árpád and the kings of the House of Plantagenet, of whom some of Shakespeare's history plays are about. The representation follows these rules: the x-axis behaves specifically. Initially, the value $x=0$ corresponds to the year 1000 (`origoYear=1000`). It is also given how many pixels a year is (`pixelsPerYear=5`). Thus, a given year (`year`) must be drawn at the following coordinate: $(year - origoYear) * pixelsPerYear$. That is, e.g. the death of St. Stephen in 1038 goes to $(1038 - 1000) * 5 = 190px$.

- a. (1 point) Draw vertical red lines from year 1000 to 1500 for every 50 years, and write the year number next to them! Initially, only 3 lines will be visible, the rest will be drawn outside of Canvas.
- b. (2 points) Draw the intervals of the reign of the kings of the House of Árpád as green rectangles!
- c. (1 point) Do the same with the English kings of the House of Plantagenet!
- d. (1 point) Write the name of the king and the beginning and end of his reign in the rectangles!
- e. (1 point) Press "Left" button to move the drawing to the right by 10 years. To do this, reduce the value of the

variable `origoYear` by 10, and redraw the Canvas.

- f. (1 point) Press "Right" to move the drawing to the left by 10 years. To do this, increase the value of the variable `origoYear` by 10, and redraw the Canvas.
- g. (1 point) Press the "Start animation" button to move the timeline continuously. You can use any timer to do this, and in an animation step, change the variable `origoYear` by `dt`. The animation does not have to be time-based.

TIMELINE



4. task: Attendance list (8 points)

An attendance list is given on the page: checkboxes in the cells of a table, some of which are already checked at the beginning.

- a. (2 points) When the page loads, run a function that gives each table cell (`<td>`), that has a checkbox, a `present` style class. *Technical help: select all the checkboxes and iterate through them!*
- b. (2 point) By clicking on a table cell, focus on its checkbox (call the `focus()` method of the checkbox)!
- c. (1 point) Clicking on a checkbox will turn it on and off, and every time recolor the table by calling the coloring function from subtask a.!
- d. (2 points) Press the four arrow keys to read the x-y coordinates of the cell containing the focused checkbox, and write that x-y coordinate to the console. If there is no focused checkbox, nothing should happen!
- e. (1 point) Press the arrow keys to move in the appropriate direction, and focus on that checkbox. That is, when you press the down key, the checkbox one cell below must be focused!

ATTENDANCE LIST

NAME	PRACTICE 01	PRACTICE 02	PRACTICE 03	PRACTICE 04	PRACTICE 05	PRACTICE 06	PRACTICE 07
Name1	☒	☒	☒	☐	☐	☐	☐
Name2	☐	☒	☒	☐	☐	☐	☐
Name3	☒	☒	☐	☐	☐	☐	☐

JavaScript évfolyam ZH angol (2021/22/1)

Criteria	Ratings		Pts
1/a	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
1/b	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
1/c	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
1/d	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
1/e	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
2/a	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
2/b	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
2/c	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
2/d	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts

2/e	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/a	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/b	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
3/c	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/d	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/e	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/f	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
3/g	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
4/a	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
4/b	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
4/c	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
4/d	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
4/e	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts

Total Points: 32.0